# Ground Truth That Does Not Ground: Substituting Reference Solvent $\sigma$-Profiles Degrades COSMO-SAC Solubility Prediction

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## Abstract

Physics-informed models route a prediction through a quantity that can be looked up, so supplying the tabulated value ought to improve them. We test that assumption for solid–liquid solubility, where a graph neural network predicts the molecular charge-density profiles a fixed COSMO-SAC layer consumes and quantum chemistry tabulates independently. Grounding names two opposite operations, and they do not behave alike. Substituting the tabulated profile for the learned one at prediction time makes solubility prediction worse at every seed, and degrades the ranking of solvents for a solute, which is the decision the model is used for. Supervising the same network against the same database during training carries no established sign. The operation that fails is the one available in practice, because it needs no retraining. Infinite-dilution activity data separate a deployed thermodynamic model's own error from what its inputs leave unresolved, and the model's error dominates on one solvent chemistry. A search over strata identifies that cell, and an independent set reproduces it at half its size without meeting this paper's own stability rule, so we report it as the instrument's output and not as an established effect. Where the fixed model is the binding constraint, a more accurate input exposes its misspecification instead of repairing it.

## 1 Introduction

How much of a solid dissolves in a given liquid decides whether a drug is formulated as a tablet or an injection, which solvent a crystallisation runs in, and how a product is purified. Each value is measured one solute, one solvent and one temperature at a time, so most industrially relevant combinations remain unmeasured, and a laboratory that needs one is usually choosing between running an experiment and trusting a prediction.

Physics-informed predictors are attractive for reasons that have little to do with accuracy, on which they usually trail. Their intermediate quantities are things chemistry already knows how to measure and tabulate, so a prediction made through them can be taken apart and audited, and each part can be supervised against reference data a direct regressor has no way to use. The fixed equation is also a prior: it asserts for free what a purely statistical model would have to learn from data it does not have, which is the situation whenever the question concerns a molecule unlike any in the training set.

That design carries an assumption so natural it is rarely written down. If the model's intermediate is a quantity someone has already tabulated, then a better value for it can simply be looked up and handed to the model, with no retraining — and the prediction should improve. The assumption is what makes the intermediate worth having: it is the difference between a physical model and an elaborate parameterisation. This paper tests it, on one such model and one such intermediate, and finds that it fails. Two further things make solubility a hard case for it. Independent determinations of the same system disagree by enough to floor what any predictor can achieve, and dissolving a solid trades breaking a crystal against mixing the freed molecules into the solvent, so a predictor must be right about two unrelated kinds of chemistry at once, on molecules unlike any it was shown.

Predictors divide by how much physical structure they impose. Direct models regress solubility from descriptors or from a representation learned off the molecular graph; FastSolv[1] does so with a deep neural network and leads on extrapolation to new solutes. They are the more accurate there and they are opaque, the prediction arriving as a single number, silent about which part of the chemistry it captured. Physics-informed models instead predict the quantities the physical picture names—an activity coefficient for the solute–solvent interaction, a melting temperature and heat of fusion for the crystal—and combine them through a fixed equation. SolProp[2] does so with a thermodynamic cycle of separately learned components. The fixed equation buys decomposability: each part is a quantity chemists tabulate on its own, so it can be supervised against reference data a direct regressor cannot use. On accuracy the black box leads and the physics-informed cycle trails it. Learned surrogates for the activity coefficient split the same way. Winter et al.[3] predict $\ln \gamma_2^\infty$ from SMILES strings, the line notation for molecular structure, with a language model; Sanchez-Medina et al.[4] and Wang et al.[5] embed a Gibbs–Helmholtz constraint in a graph network for its temperature dependence. All three make the activity coefficient the regression target. A second line keeps a mechanistic model and learns its *input*. COSMO-SAC[6] is a segment-activity reformulation of Klamt's conductor-like screening model for real solvents (COSMO-RS), and the $\sigma$-profile it consumes—the distribution of screening-charge density over a molecule's cavity surface—is that framework's construct, tabulated from quantum chemistry.[7] Predicting it from structure removes that calculation from the pipeline. Group-contribution[8,9] and then learned predictors[10–12] have supplied it in place of the calculation. How far the tabulated reference itself moves with the quantum chemistry and solvation treatment behind it is its own question.[13] Islam and Chen[14] instead regress an *apparent* profile from phase-equilibrium data, the pre-neural form of the effective profile this work measures. TeNNet-SAC[15] does so while also learning the segment-activity kernel and the cavity geometry, and reports accuracy better than COSMO-SAC's after end-to-end finetuning. Its headline is the opposite of this one's, and the two score different operations. Its gain is a training-time gain, which over five leak-free seeds has no established sign here. Its externally supervised profile predictor then stays frozen while only the output head is finetuned, so the closure's gradient never reaches the intermediate. This work's negative result is about the inference-time substitution, which TeNNet-SAC leaves unreported, on an intermediate trained through the closure with neither safeguard.

A third design removes the intermediate. HANNA[16] parameterises the excess Gibbs energy directly from SMILES with no quantum chemistry, hard-coding the pure-component limit, the Gibbs–Duhem relation and permutation symmetry in the architecture rather than the loss. It reports higher accuracy than UNIFAC,[17] the group-contribution activity-coefficient method standard in process design, over 317,421 mixture points. The two published designs bracket what this work measures: admissibility can be bought with quantum labels or built into the functional form, and this model does neither.

When a fixed physical structure improves or degrades a learned model is the broader question of physics-informed machine learning.[18] Its dominant forms impose structure as a differentiable object: a governing equation enforced through the loss,[19] gray-box hybrids composing learned components with fixed mechanistic maps,[20] and operator learning with a trainable correction to a misspecified equation.[21] More physics is expected to improve generalisation, and the published evidence is inconsistent in direction: the same structure has documented failure modes in which it makes optimisation or accuracy worse.[22] In neural solvers for partial differential equations a learnable term added to a fixed discretised operator absorbs truncation artifacts instead of physics, so the apparent interpretability is illusory.[23] The error source there is numerical, which puts it beyond external reference. The same concern appears in *concept-bottleneck models*,[24] networks routed through a layer of named, human-interpretable concepts that are the whole of what the prediction is made from. Concept bottlenecks have the shape of a learned intermediate feeding a fixed physical map, and there the failure is *concept leakage*: a learned concept silently encodes information beyond its nominal meaning.[25–27] One route to it runs through the downstream model: a misspecified head forces the encoder to deform the concept into carrying a dependence beyond that head's reach.[28] Overwriting a frozen head's leaked concepts with ground truth can then degrade accuracy.[29] The penalty is older than that literature. Koh et al.[24] already report it: in a joint bottleneck trained to prioritise label accuracy over concept accuracy, replacing the predicted concepts with the true ones at test time slightly increases test error. The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O'Hagan[30] and its calibration critique.[31] Four further literatures—forecast decomposition, quasi-maximum likelihood, identifiability, and the validity of composed pipelines—supply pieces we use below.

The design rests on an assumption rarely stated because it is taken for self-evident: a model supplied with physical inputs closer to their true values predicts better. It needs testing, and for one class of intermediate it can be tested outright, because a reference value is tabulated and can be handed to the model in place of the one it computed. When a predictor of this shape is inaccurate, either the fixed physical model is wrong or the learned quantities feeding it are uninformative, and the two call for opposite remedies. The literature lacks a way to *measure*, rather than diagnose, which of the two limits accuracy, and whether the learned intermediate has stopped reporting the quantity it is named after. In the concept-bottleneck case the external reference that would settle either question is missing. We supply one—a reference $\sigma$-profile for solubility, tabulated substituent constants for a second chemical domain—and hold the mechanistic model fixed, so the tabulated value stays available as an external reference for the learned one.

Here a graph neural network predicts the pair of charge-density distributions COSMO-SAC consumes, in place of the quantum-chemical calculation that ordinarily supplies them. The tabulated counterpart enters at either of two points: as a target the network is trained toward, or as a replacement for what the network computed at the moment of prediction (Fig. 1). We use *grounding* for those two operations on tabulated physical values inside a hybrid model, and not in the sense the word carries in computer vision or language processing. Of the two, only one carries a sign. Training toward the reference moves the error four ways out of five and the fifth the other way; supplying it at the moment of prediction raises the error at every seed. Tabulated profiles serve a third purpose on activity-coefficient data, where they divide the error of a fixed model between the model itself and the inputs handed to it. Solubility withholds that division, a solubility measurement constraining the crystal and the mixing contributions jointly and leaving the blame between them unassigned. The division returns a map and not a verdict: one answer per stratum, a stratum being a chemically defined set of rows. The chemistry the model is asked about decides whether a fixed model's own error stands above what its inputs fail to carry. In the strata where it does stand above, a more accurate input exposes the model's bias instead of removing it. The learned distribution meanwhile drifts from its tabulated counterpart along a direction shared across molecules, and stops reporting the quantity it is named after. Tabulated substituent constants behave the same way inside a fixed Hammett relation, the linear equation predicting how readily an acid gives up its proton from the constants tabulated for its substituents. We use the VT-2005 profiles—the Virginia Tech 2005 tabulation[7]—not as ground truth for solubility prediction as such, but as an external physical reference against which to diagnose whether the learned intermediate keeps its physical meaning or has drifted to compensate for the closure's errors.

The article is organised as follows. Section 2 defines the composed predictor, the decomposition of its error into a part belonging to the fixed physical model and a part belonging to the inputs handed to it, and the data, training and evaluation protocols. Section 3 reports the two grounding operations on solubility, what substitution costs in accuracy and in the ranking of solvents, how the learned intermediate departs from its tabulated counterpart, and the error decomposition on activity-coefficient data for two fixed models. Section 4 states the result and its scope.

This work does not propose a solubility model and claims no accuracy for the one it uses. The model is the instrument in which a property of a fixed thermodynamic closure is measured, and every result below is a contrast between two readings of one trained network rather than a comparison between networks.

# 2 Computational methods

## 2.1 The composed predictor and its error split

For a solute $s$ in solvent $v$ at temperature $T$, the target is $y = \ln x_2$, the log saturated mole fraction. Dissolving a solid trades breaking the crystal lattice ($\Phi$) against mixing the freed molecules into the solvent ($\ln \gamma_2$), and solid–liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = -\underbrace{\frac{\Delta H_{\text{fus}}}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right)}_{\Phi(T)\ \text{(crystal)}} - \underbrace{\ln \gamma_2(x_2, T)}_{\text{activity}}, \quad (1)$$

with no constant-$\Delta C_p$ correction ($\Delta C_p$ being the difference in heat capacity between the liquid and the solid). That term is absent from the displayed equation because it is absent from the deployed model, which fixes $\Delta C_p = 0$, the head that would predict it being off in the pinned configuration. The exposure is one-signed and substantial. At the group-contribution value this corpus carries, $60\ \mathrm{J\,mol^{-1}\,K^{-1}}$, the omitted term is a median 0.32 in $\ln x_2$ over the 1318 labelled test rows carrying a measured $T_m$, and positive on every one of their 31 solutes. That is the size of the substitution penalty of §3.1. It cancels exactly in the evaluation-time substitution, where $\Phi$ is common to the two arms, and §S1 gives where else it acts. The activity term is produced by a *differentiable closure* $g$. A graph encoder predicts a $\sigma$-profile (the distribution of screening charge density over a molecule, the histogram a COSMO model consumes) for each molecule, and COSMO-SAC,[6] a segment-activity reformulation of COSMO-RS, maps the pair of profiles to $\ln \gamma_2$. The segment-interaction energy inside that map is the closure's *kernel*: the electrostatic-misfit and hydrogen-bonding terms setting what two surface segments cost each other, and the part the published revisions rewrite. A closure variant is therefore named for its kernel's year — the *2002 kernel* this pipeline runs, and the *2010 kernel* that types segments as hydrogen-bond donors or acceptors instead of treating hydrogen bonding with one parameter. Write $z^\star$ for the *reference* latent, a tabulated $\sigma$-profile, and $\hat{z}$ for the head's own prediction. **Two profile databases supply $z^\star$ below and they are never interchangeable.** *VT-2005* is the Virginia Tech 2005 tabulation:[7] the $\sigma$ head is supervised against it, and it is what replaces $\hat{z}$ at prediction time. *UD* is the University of Delaware successor,[32] redistributed with the NIST reference implementation,[33] and the activity-coefficient rows of §3.4.1 are matched

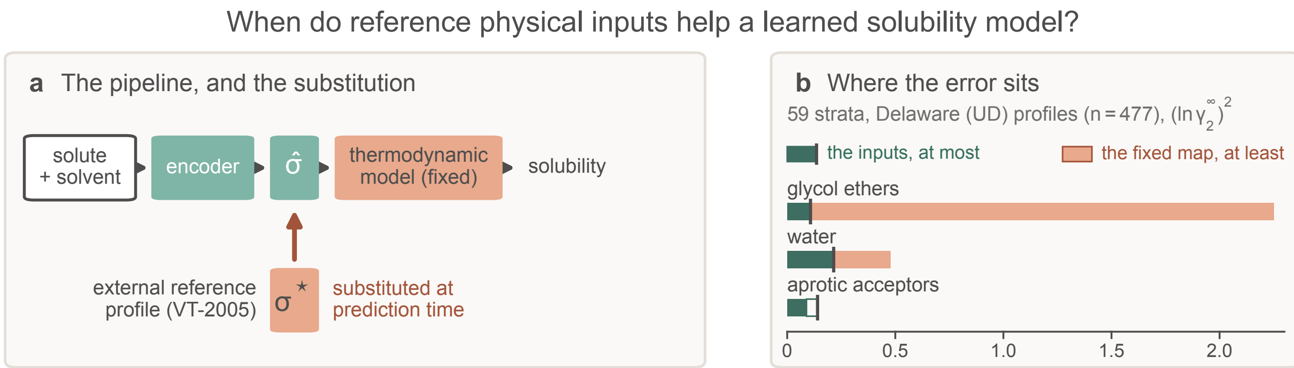


Figure 1: **Overview.** **(a)** A learned encoder predicts a charge-density profile $\hat{\sigma}$ that a fixed thermodynamic model turns into a solubility, and the arrow marks the substitution of the reference $\sigma^\star$ for it. Training-time supervision is omitted. **(b)** The counterpart error on infinite-dilution activity data, for three of the fifteen strata that carry a bound, out of fifty-nine searched: most of the map supports no statement at all. Each block is a one-sided bound, not an interval, so what the panel shows is that the input bound moves two-fold across the three classes while the total error moves twenty-five-fold. Of the three, only the glycol ethers meet the admissibility rule; water's sign does not survive dropping a contributing laboratory, and for the aprotic acceptors (solvents with no O–H or N–H) the bound runs past the total error, so neither is established. Table S18 gives all three verdicts with their tests.

to it. Every set, margin and figure below names which it is computed on, and we read no quantity across them; the one place they are deliberately set against each other is the database-to-database bracket of §3.4.3. Feeding $z^\star$ in place of $\hat{z}$ (a $\sigma$-oracle) isolates the closure from encoder error: were the closure well specified, $g(z^\star)$ would be the best available activity estimate. Each trained configuration, and each closure variant evaluated, is an *arm*; a *Direct-GNN* control arm drops the closure and emits $\ln x_2$ directly, and we tuned the two separately (§2.4).

We state everything below for a general *composed predictor* $\hat{y} = g(h(x))$ in which a fixed closure $g$ consumes a learned physical intermediate $z = h(x)$, against a matched free predictor that shares the encoder and drops $g$. Solubility is one instance, and a second is the closure of §3.5 for p$K_a$, the acid dissociation constant on a negative logarithmic scale.

Solubility alone underdetermines the crystal/activity split of Eq. (1). The deficiency closes under a rank-completing external constraint and nothing weaker — a direct crystal label, or a fixed activity slope (Lemma S1, §S2). That is why an under-determined factor needs external single-component data and why we expect compensation between the factors, and it is silent on whether grounding helps.

#### 2.1.1 The closure–insufficiency decomposition

Let $m$ be the closure's target (here the experimental infinite-dilution activity coefficient (IDAC) $\ln \gamma_2^\infty$, in §3.5 the measured p$K_a$) and $g(z^\star)$ the closure evaluated on the reference latent.

**Lemma 1** (Exact closure–insufficiency decomposition and its one-sided bounds)**.** *Let $g$ be a fixed (non-fitted) closure and let $z^\star$ denote its* entire *argument (the profile pair together with the temperature wherever the closure reads one). Then* (a) *the reference-input mean squared error splits exactly,*

$$\underbrace{\mathbb{E}\big[(m - g(z^\star))^2\big]}_{\textit{ref.-input MSE}} = \underbrace{\mathbb{E}\big[\mathrm{Var}(m \mid z^\star)\big]}_{B_{\text{insuff}}\ (\textit{irreducible})} + \underbrace{\mathbb{E}\big[(\mathbb{E}[m \mid z^\star] - g(z^\star))^2\big]}_{B_{\text{closure}}\ (\textit{decoder bias})}, \tag{2}$$

*the cross term vanishing identically;* (b) $B_{\text{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$*;* (c) $B_{\text{insuff}} \leq$

$\mathbb{E}[\mathrm{Var}(m \mid \mathrm{bin}(g(z^\star)))]$ *for any binning of* $g(z^\star)$*; and* (d) $B_{\mathrm{insuff}}$ *is a functional of the law of* $(m, z^\star)$ *alone and does not involve* $g$*, so it is identical across closure conventions, while the bound in (c) is not.*

The proof is in §S2. Two of its parts are conditions of use rather than steps, and both are stated here. **The identity holds because $g(z^\star)$ is $\sigma(z^\star)$-measurable, and it fails if any argument of $g$ is left out of the conditioning variable**: with $g = g(z^\star, T)$ and $T$ excluded, the cross term need not vanish and the binning stops being a coarsening of the conditioning $\sigma$-algebra, which takes the identity and the upper bound with it. Where $g$ reads a temperature (the broad IDAC set of §3.4.1, dim = 103) we accordingly take $z^\star = (\text{profile pair}, T)$. On the VT-2005-matched set every row sits at 298 K and the two readings coincide. The Hammett closure of §3.5 reads a per-series constant pair as well as the substituent constant, so there $z^\star = (s, \sigma_{\mathrm{H}}^\star)$ and we condition on both.

$B_{\mathrm{closure}}$ is the systematic discrepancy of a fixed physical map in the sense of Kennedy–O'Hagan,[30,31] and its dominance over $B_{\mathrm{insuff}}$ is the empirical content of the claim that the ceiling is set by the closure and not by the inputs. **$B_{\mathrm{closure}}$ is the error of a composite object.** The identity compares $g(z^\star)$ against $\mathbb{E}[m \mid z^\star]$, and $z^\star$ is produced by a procedure: for $\sigma$-profiles, a quantum-chemical calculation on one conformer and one protonation state per molecule at one level of theory. $B_{\mathrm{closure}}$ is therefore a joint property of the map and of that procedure, not of the kernel's functional form, and a displaced reference loads onto it through the curvature of $g$ even where $g$ is exactly specified (§S8). The deployed pipeline consumes that same composite, so every attribution below is to the pair. Where model-discrepancy calibration *fits* a correction from held-out observations, here $g$ is fixed and $z^\star$ supplied from outside the fit, so we *attribute* the composite's error instead, which is available only where an external latent oracle exists. §S2 gives the conditions this exactness depends on and what fails without them.

Because $B_{\mathrm{insuff}}$ is bounded above and $B_{\mathrm{closure}}$ below, a positive *separation margin* MSE $- 2B_{\mathrm{insuff}}^{\mathrm{up}}$ establishes that the closure's own error exceeds what input insufficiency can account for. A margin at or below zero fails to separate and licenses nothing. **Its size is a lower bound and not an estimate**: every margin below says the closure's error exceeds its inputs' by *at least* the number printed, and the design cannot say by how much more. Over-estimating $B_{\mathrm{insuff}}$ therefore produces no artifact in what is reported, and a low cluster-bootstrap frequency is loss of resolution rather than evidence for the inputs (§S2).

The decomposition is the measurement the conclusions rest on: it is assumption-light, needing neither a representable encoder nor a lossless bottleneck. Beside it we report the directly observable *grounding gap*, which certifies $B_{\mathrm{closure}} > 0$ only under two assumptions this work cannot verify, so we read it as corroborating evidence and rest the model-dominance conclusion on the decomposition (§S6.9).

## 2.2 Data curation: sources, splits, and labels

Solubility measurements come from BigSolDB 2.0:[34] 108,287 rows over 1525 solutes and 212 solvents. Melting points merged in as single-component crystal labels bring the modelled corpus to 127,930 rows over 20,766 solutes and 228 solvents between 243 and 438 K. We partition by Bemis–Murcko scaffold[35] into 111,724 training, 8,103 validation and 8,103 test rows, so that every test solute is unseen at training. A melting-point-independent miscibility/structure filter sets row membership, and we merge on crystal $T_m$, $\Delta H_{\mathrm{fus}}$, Hansen solubility parameters and infinite-dilution activity coefficients $\ln \gamma_2^\infty$ as per-solute labels where available. The rows of a split carry different labels: of the 8,103 test rows, 5608 carry a solubility measurement while the other 2495 carry a melting point alone, so every accuracy number below is on 5608 rows and says so (§2.4). 796 of those 5608 rows, 14.2% of them, carried by 26 of the 147 test solutes, have a solute of more than one component (salts, two ionic liquids and four hydrates). For those, the solid in equilibrium with the saturated solution differs from the one-component crystal Eq. 1's ideal term is written for, and all 26 of those solutes lack a measured melting point. They stay in: dropping the stratum moves the control's mean and the grounded arm's the same way and widens the physics penalty this paper reports, so removing it at this stage would flatter the comparison (§S3.2). Solid form is uncontrolled on both label sources, the solubility rows aggregating literature measurements on whatever form each laboratory had, and the $T_m/\Delta H_{\mathrm{fus}}$ labels being merged from a separate compilation free to refer to another form. A polymorph difference therefore enters the ideal term as a systematic and the inter-laboratory disagreement below as part of its spread, and both documents leave it unseparated. Crystal supervision is scarce: the test and validation splits contain essentially no measured $\Delta H_{\mathrm{fus}}$, the quantitative basis of the identifiability argument (§2.1). The external single-component data we use for grounding are an open crystal-property pool (∼15,000 melting points), the VT-2005 $\sigma$-profile database, and ThermoML infinite-dilution activity coefficients.[36] Two hazards the pipeline handles explicitly (melting-point unit detection, and the instability of the seeded scaffold split across pipeline versions) are recorded in §S8.

**The label-noise floor, and the two functionals it is read with.** Independent determinations of the same system disagree, and that disagreement floors what any predictor can achieve. Grouping the raw BigSolDB rows by (solute, solvent, temperature to 1 K) and keeping those with at least two distinct sources, the mean absolute disagreement is 0.15 $\ln x_2$ units over the 3948 two-source groups and 0.31 over the 349 three-source ones. The scaffold-split MAE reported here (1.70 to 1.85) exceeds the larger of those by a factor of five, so model error dominates label noise. The published aleatoric estimate for this task is a different functional on a different base[1] and we keep the two apart, pooling and converting neither into the other; §S4 gives the other functionals of this floor and the comparison.

**Disclosure: the scaffold guarantee is not certified for these runs.** The released stream builder excludes any pool molecule whose scaffold appears in validation or test. One $\sigma$-stream build on disk was nevertheless generated against the wrong split files, and per-run stream provenance went unlogged, so which build fed which arm lies beyond the artifacts' reach and the guarantee is a property of the builder rather than of these runs. Of the 147 solutes spanned by the 5608 labelled test rows, 7 carry their own reference profile in that build, and 8, those 7 among them, carry a scaffold relative. The second count is therefore the union, and deleting it deletes both. The ungrounded arm carries no $\sigma$ stream either, so the supervision contrast is the one a leak can reach (§3.1). The file evidence, the contrast each fact bounds, and the one molecule the leak touches in the $n$=44 analysis of §S6.10 are in §S8.

**The leak-free re-run.** We fixed its protocol before its numbers exist (§S8). The ungrounded and the grounded arm run at five seeds (42 through 46) on the published COSMO-SAC configuration unchanged, on a $\sigma$ stream rebuilt against *this* split's files and certified inside each training run. Both are read on the same 5608 labelled test rows and reported seed by seed. We fixed the decision it settles in the same pass, stating it in three branches rather than in the favourable one alone (gains of one sign, gains straddling zero, gains of the opposite sign), together with the list of displays the five-seed values replace. The supervision contrast is what that decision hangs on. All three branches leave the substitution result standing: it is one checkpoint evaluated two ways, beyond the reach of any $\sigma$-stream build.

## 2.3 Model, closure, and training

What follows specifies the instrument rather than proposing a method. None of the architecture, the closures or the training schedule is offered as a contribution, and no result in this paper turns on a choice among them; they are described in the detail they are because a measurement of a fixed closure's error is only as reproducible as the pipeline it was taken in. A shared message-passing encoder embeds solute and solvent, a bipartite cross-attention block lets them interact, and the resulting pair representation feeds every head (§S1). We implemented two further backbones and both stay outside the numbers reported here; the physics-channelled one was run against this encoder and trailed it, an observation recorded with no set, seed count or interval attached (§S1), so we claim no accuracy comparison between backbones. A fusion head predicts $T_m$ and $\Delta H_{\text{fus}}$, giving the ideal term $\Phi(T)$ of Eq. (1). The activity term $\ln \gamma_2$ comes from one of three differentiable closures. Two are controls: the fitted NRTL (non-random two-liquid) excess-Gibbs-energy model, and its symmetric one-parameter collapse. The third is the closure under study and is parameter-free: a $\sigma$-profile head emitting a 51-bin screening-charge-density profile per molecule, and a COSMO-SAC layer mapping the profile pair to $\ln \gamma_2$.

External single-component data enter through *grounding streams* (the training-time sense of grounding). Each stream is a sidecar of *self-solvent* rows (no solubility label, only the target factor's mask active) interleaved into training under the guard above. It grounds the crystal factor from the melting-point pool and the activity factor from the $\sigma$-profile database, and so identifies from external data a factor that solubility alone leaves non-identifiable and that a black-box predictor is unable to use. Training runs a three-stage curriculum under a weighted loss, and the stages are named by number wherever one of them is referred to below: *P1*, a warm-up on the auxiliary streams alone; *P2*, joint training against solubility; *P3*, a low-rate stabilisation stage. The schedule pinned for the arms of Fig. 2 freezes the $\sigma$ head at the end of P1, so the $\sigma$ stream enters the warm-up only.

**The in-house closures, and where their validation stops.** The dispersion-off closures in the fidelity measurement (§3.4.3) are our own differentiable implementations, so those arms share one code path, and each consumes its own model-prescribed $\sigma$-averaging (Mullins *et al.* for 2002,[7] Hsieh *et al.* for 2010[37]). Our 2002 layer (the *2002, all channels* arm wherever the kernels are compared, so named because it sums the segment contributions of all three profile channels rather than the nonpolar one alone) reproduces the NIST reference COSMO-SAC-2002[33] to RMSE 0.0035 $\ln \gamma$ on identical VT-2005 profiles. Our COSMO-SAC-2010/dsp layer,[37,38] built on the *typed* latent of three donor/acceptor-resolved $\sigma$-profiles per molecule in place of one, reproduces the NIST 2010 reference with dispersion *off* to RMSE $1.6 \times 10^{-4}$ $\ln \gamma$ over 400 pairs, the largest single departure being $1.7 \times 10^{-3}$ (§S1). The dispersion term is *not* validated to that standard, so we compute every dispersion-off contrast on our layers and quote the *2010, dispersion on* runs from `cCOSMO`, the C++ library that is that reference implementation[33] and not a variant of the model.

**The same closure against a published evaluation.** Reproducing a reference implementation fixes the code, and it leaves open whether the error this paper decomposes is the error a published COSMO-SAC has or something peculiar to this data assembly. Table S27 answers that: scored over the same records as a published evaluation of the same closure,[39] our differentiable layer lands inside the bracket that evaluation spans across profile databases, under both conventions. Two limits stay open: the profile database is the one axis we cannot match, and whether its 6977 points are the rows of our 7889 is unproven. Subject to them, the closure error decomposed in §3.4 is, in average absolute deviation (AAD), the error a published COSMO-SAC has. That evaluation also scores a donor/acceptor-typed later kernel on those records at about half the deployed closure's AAD. Every ceiling this paper reports is therefore this kernel's ceiling and not COSMO-SAC's, and the results below are substitutions inside a fixed map and not claims that the map is the best available.

## 2.4 Evaluation protocol, and the DirectGNN control

In the *oracle substitution* we replace the predicted $\sigma$-profile $\hat{z}$ with the reference VT-2005 profile $z^\star$ before the closure, matching by canonical SMILES. One object is replaced, not two. The $\sigma$-head emits a normalised 51-bin shape and a cavity area whose product is the area-weighted profile the closure reads, and the tabulated pair stands in the same relation. The reference area therefore reaches the residual term's $A_2/a_{\text{eff}}$ prefactor, in which $A_2$ is the solute's cavity surface area and $a_{\text{eff}}$ the closure's effective segment area, the constant patch size its segment-activity sum is written over, as part of that single replacement (§S1).

A property of our head bears on what that substitution does, and it is a defect rather than a choice. The encoder carries a separate learned adapter for the solute and the solvent role, so one molecule has *two* predicted profiles — and a $\sigma$-profile is a property of a molecule alone. The model therefore violates an identity no activity model may violate: on a self-pair $\ln\gamma_2$ must be exactly zero, and over the 2532 testable test solutes its median $|\ln\gamma_2|$ is 0.154. **No scored row is a self-pair.** On a pair of distinct molecules each profile is read in exactly one role, so no role discrepancy enters the residual bracket; the labelled and unlabelled test rows, the validation split and both activity sets contain no self-pair at all. The substitution therefore changes the profile values alone on every row this paper reports. Tying the roles remains the repair for the identity itself, which is a retraining (§S1).

We match solute and solvent independently. The substitution replaces those two profiles and stops there, the crystal head reading a solute-only embedding it leaves untouched. The bounded correction does read the closure's own parameters, so the crystal values the solver finally receives shift by a little: pooled over the five seeds, at most $3.2\times10^{-5}$ in $\Phi$ against a median $|\Phi|$ of 4.5, on the 8066 of the 8103 test rows where the substitution applied and on the 5608 labelled rows alike (§S1). The reference table covers 99.3% of the labelled test rows on the solvent side and 5.3% on the solute side, so the solvent profile carries the substitution contrast below.

We apply it three ways to rule out implementation artifacts, differing in *when* the replacement is made: at evaluation alone; injected at training so the model co-adapts; and a channel swap. The last is injected at training under a coordinate-descent schedule that freezes the crystal branch during the SLE phase, so the activity branch alone refits against it. The last two are at seed 42. An analogous override substitutes reference crystal targets. The activity target for the decomposition is the infinite-dilution $\ln\gamma_2^\infty = \lim_{x_2\to0}\ln\gamma_2$, on two sets that differ in their reference database and their regime (§3.4.1–§3.4.2).

DirectGNN shares the encoder, interaction, readout and pair representation, adds a thermometer (cumulative-threshold) temperature encoding, and maps directly to $\ln x_2$ with no heads, closure or solver, so it sees $T$ explicitly where the physics arm sees it through $\Phi(T)$ and the closure. We tuned the optimisation settings per arm, and they differ by up to an order of magnitude. The TGNNSolv$-$DirectGNN gap is therefore a comparison of two separately tuned pipelines that differ in the thermodynamic bottleneck, and not an isolation of it; every physics-penalty comparison below carries that caveat (§3.6).

We report the controlled comparisons of §3.1 as mean±sd on a fixed row set, at five seeds for the arms the leak-free re-run trains and three for the rest. That lock is the 5608 test rows carrying a solubility measurement, 5608 of the 8103 (§2.2), so all arms score identical rows. The 2495 rows outside it carry a melting point alone and are unscorable in every arm alike, the control included, each dropped for its label and not for anything an arm predicted. Which rows carry a label fixes the set: the lock's finite-prediction clause is inert, removing no row from any arm at any seed, and §S3.2 gives the decomposition. The runs behind every number reported here fall into twelve families, a family being a set of runs sharing a script, a corpus and a seed policy, and Table S29 gives each number its family, its seed count and the artifact it regenerates from.

## 2.5 Estimating the split: one cell, one convention, one rule

The instrument is a procedure taking three inputs and no others: a fixed closure $g$, a table of reference values $z^\star$ its argument can be read at, and measurements $m$ of its target on rows where both exist. Per chemically defined row set it returns the closure's own error MSE, an upper bound $B_{\text{insuff}}$ on what the inputs leave unresolved, the margin between them, a cluster interval on that margin, and a verdict of admissible or otherwise. Every margin in this paper is the binning bound of Lemma 1(c); the alternative bound of part (b) is strictly slacker on both sets and is never used (§S1.1).

Four analyst choices set that bound — the bin count, where the equal-count bin edges fall, the within-bin variance convention and the combinatorial convention — and one setting of them, with the row or the molecule pair as unit, is an *estimator cell*. All four were fixed before any margin was seen, a data-set contract in the sense Nael et al.[40] argue for. The cell is eight equal-count bins of $g(z^\star)$, the unbiased within-bin variance, the row unit and the residual-only convention, applied identically to every set and stratum whatever its size. The bin count has the most leverage of the four: it takes the broad set's aggregate margin from +0.01 to +0.96. No finding therefore rests on the cell alone — the same sweep is run for every stratum the rule passes and reported with it — and one convention governs both sets rather than the favourable one on each, which costs the broad set's margin about half its value.

Admissibility asks three things. *(a)* The stratum must be *boundable* at that cell: $n \geq 40$, eight bins holding five rows. *(b)* Its margin must keep its sign when each contributing source publication is deleted in turn, in all four unit × convention cells; that clause is load-bearing, since dropping it lets the set taken whole pass. *(c)* Admissible cells holding identical rows count once. §S1.1 states all four rules in full, with the class cascade, the demotion test and what each of them costs.

# 3 Results and discussion

Every claim below carries its set, its size and its scope in the sentence that makes it, and §S3.1 collects the same nineteen side by side as one table (Table S3). Two reading conventions travel with them. A cluster-bootstrap frequency $P$ is the frequency, over resamples, with which the margin it is attached to stays positive — a resampling frequency and *not* a $p$-value against any null. And every interval below is a percentile interval over the unit named beside it. §S1.2 gives those units, the draw counts, the rounding rule the counts support, and the two spans that are of a different kind.

## 3.1 Supervising the intermediate toward the reference, and substituting it at prediction time

The word "grounding" names two opposite operations on the same VT-2005 database. Every solubility figure in this subsection and the next is an $\ln x_2$ MAE over one row set, the 5608 of the 8103 scaffold-split test rows that carry a solubility measurement (Table S1). The single-seed control arms below carry that identical set, so every comparison here stays inside one row set. Supervising the $\sigma$ head against the database *during training* moves solubility MAE from 2.037±0.064 (ungrounded) to 1.927±0.094 (grounded) over five leak-free seeds. The mean of +0.11 is not a sign. The per-seed gains are +0.149, +0.122, +0.175, −0.017 and +0.122, the two arms' per-seed values interleave, and at seed 45 the grounded arm is the worse of the two. The two arms differ in training budget. The grounded arm precedes the shared curriculum with an early-stopped $\sigma$ warm-up on the grounding pool; the ungrounded arm sets that budget to zero (§S3.2). We left unrun the warm-up against permuted $\sigma$ targets that would separate the two, so the gain prices the intervention entire, extra pretraining and accurate targets together, rather than the accuracy of the reference values by itself. Those spreads are over initialisation and batch ordering, so they price the training run rather than the draw of test solutes. Pooling the five seeds and resampling the 147 test solute clusters, we put the gain at +0.110, 95% percentile interval $[+0.001, +0.222]$. The two prices hold different things fixed and they do not agree: over the draw of solutes the interval clears zero at its endpoint, and over training runs the gain has no sign at all. The ungrounded arm carries no $\sigma$ stream, so this is the paper's only contrast a $\sigma$-stream leak could inflate, and it is what the re-run was pre-committed to settle. Its stream is certified inside each training run and reaches no test molecule: 0 by canonical SMILES against 5101 held-out SMILES, and 0 by Bemis–Murcko scaffold against 2702 held-out scaffolds (§S8). On these arms there is nothing to delete. The published three-seed family, whose stream build is uncertified, bounds where a leak can act *there*, and is what limitation (xii) refers to (§S6.8). The leak-free build supersedes it: in that build a leak cannot act at all. Substituting the reference profile for the learned one *at inference* degrades it, and that substitution alone is the effect this paper is named for. It erases any positive $R^2$ (Fig. 2): the model predicts solubility better *from its own learned $\sigma$-profile* than when *given the external reference one*, at MAE 1.93 against 2.34 in $\ln x_2$ units. The gap holds over the same five seeds, and the two arms' per-seed values are again disjoint. Resampling the same 147 solute clusters over the pooled five seeds, we put that penalty at +0.408, 95% percentile interval $[+0.29, +0.54]$. The median per-row gap is +0.30 against the mean's +0.41, so the effect sits at the middle of the distribution and grows heavier above it. Of the figure's eight arms, two are non-COSMO controls, DirectGNN and a fitted NRTL closure. Five COSMO-SAC arms differ in how the $\sigma$ head is trained (the warm-up asymmetry above included) and in what it is fed. The sixth (*grounded+comb.*) differs instead in the closure, restoring the Staverman–Guggenheim combinatorial term. *Ref. $\sigma$ (eval)* is the grounded checkpoint re-evaluated with the reference profile substituted at test time, and *ref. $\sigma$ (train)* the co-adaptation control injecting it during training instead.

The +0.41 penalty is the evaluation-only substitution, and part of it may be unrelated to the closure: swapping an input distribution at test time degrades a pipeline on its own. Separating that component needs the reference profile injected *at training*, so that the model co-adapts to it, and no such arm was run at five seeds. The share of the substitution gap that co-adaptation would carry is therefore unmeasured here, and the penalty below is the two operations together (§S3). The five-seed arms are the ones that can be set against one another, and there the evaluation-only substitution costs several times what supervision gains (+0.41 against −0.11). Both carry the co-adaptation confound, so that ratio measures something other than a co-adaptation-free magnitude, and this work supplies no arm that does. The −0.11 has no sign of its own either: its per-seed values are +0.149, +0.122, +0.175, −0.017 and +0.122. The ordering of the two operations is therefore unresolved, and only their signs are carried forward.

## 3.2 What the substitution costs: level, spread, and the choice of solvent

Figure S8 sets the same rows out predicted against measured, for the control and three of the closure arms. Two things are visible there that the arm ordering hides. First, every arm is severely shrunk toward the corpus mean. The ordinary least-squares slope of predicted on measured is 0.515 for DirectGNN, 0.399 ungrounded, 0.511 grounded and 0.658 with the reference profile, all of them well short of 1. In every panel the median trace crosses the identity: sparingly soluble compounds are called more soluble than they are and highly soluble ones less. All four slopes are seed 42's, the figure's seed, with one run behind each. The shrinkage originates outside the closure: at that seed the closure-free control's slope differs from the grounded closure's by 0.004, so whatever produces it lives in the encoder and the data. Second, the reference-profile arm is the *least* shrunk and the worst. Its slope is the closest of

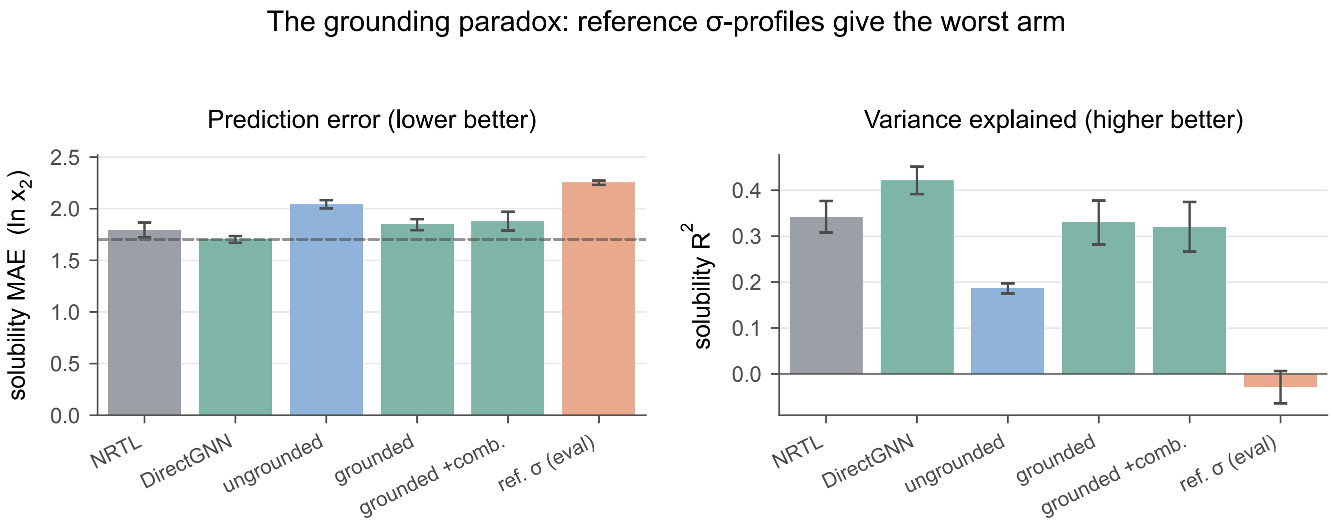


Figure 2: Controlled comparison on the 5608 labelled test rows of the scaffold split. Bars are five seeds, mean ± a population standard deviation over them. The ungrounded, grounded and $\sigma$-oracle arms come from the five-seed leak-free re-run. Dashed in the left panel alone, the DirectGNN reference.

the four to 1 and its prediction spread exceeds the measurement's, yet its MAE is the highest and its $R^2$ is indistinguishable from zero. Substituting the reference restores spread that is misaligned with the truth, and its worst region is the sparingly-soluble tail it appears to reach. The closure is driven hardest in that tail. At seed 42 the substituted arm returns 119 rows with $\ln \gamma_2 > 10$ where the learned arm returns none, the learned arm's largest being 7.77 against the substituted arm's 14.6. And 141 of its $\ln x_2$ predictions fall below $-15$, to a minimum of $-22.3$ where the corpus itself stops at $-17.7$. The worst 500 of the 5608 rows carry 0.25 of that seed's +0.26 mean gap. Both arms are scored at the 30 segment iterations of §2.3, and the rows the closure is driven hardest on are also the ones the truncated fixed point is least certified on. The count is a lever: on the one trained head the deposit retains, the substitution penalty moves fifteen-fold with it and in the other direction, which prices the lever and not this paper's +0.41 (§S6.8). The restored, misaligned spread is the misspecified map made visible on the solubility axis.

The substitution also costs the decision the model is used for. The separate-tuning confound of §3.6 leaves this comparison clear: the reference-profile arm re-runs the grounded checkpoint's own weights over the same rows and changes exactly one thing, the profile fed to the closure. One tuning serves both sides. They differ in $\Phi$ by at most $3.2 \times 10^{-5}$ over these 5608 rows at the worst of the five seeds. Co-adaptation is the confound that survives. Like the MAE gap above, this is an input swapped at evaluation; we ranked no arm that had trained on the reference profile, and what co-adaptation would do to an ordering stays unmeasured here. We group the labelled test rows by solute and temperature and rank the solvents inside each group (823 groups over 107 solutes; Table S22). The mean per-solute Spearman correlation then falls, pooled over the five seeds, by $-0.185$ $[-0.27, -0.11]$, every per-seed interval excluding zero. A rank correlation is defined for both arms in 819 of those groups, and Table S22 gives the arms seed by seed. Four of the five falls exceed any difference between seed replicates of a single arm on this measure, the largest of which is 0.132. The fifth, at seed 42, is 0.114 and does not: at that seed the fall sits inside the spread one arm shows between its own seeds. The deposit carries that benchmark for the rank correlation alone, so we read it against that measure and leave the two below unbenchmarked. Every interval here is a 95% percentile bootstrap over those 107 solute clusters, the effective sample rather than the 823 groups, and is printed to two decimals because a third would be a property of the bootstrap seed (§S3.8). Normalised discounted cumulative gain over the top three solvents (nDCG@3) falls by $-0.071$ $[-0.11, -0.03]$, and top-1 accuracy by $-0.088$ $[-0.17, -0.01]$. We quote that last figure pooled, and pooled only: three of its five per-seed intervals span zero, so of the three ranking metrics it is the one whose sign the per-seed columns leave open. In 42 to 53% of groups the model selects a different solvent, and asymmetrically. The learned profile by itself picks the measured best solvent in 18 to 24% of groups, the reference profile by itself in 10 to 15%. All of that is an ordering rather than a level. A separate affine recalibration fitted inside every group removes 91% of the MAE gap, $+0.426$ $[+0.30, +0.55]$ to $+0.040$ $[-0.00, +0.08]$. The map carries a slope as well as an intercept, the largest concession a miscalibration reading can ask for. What is left of the gap no longer stands clear of zero. A per-group map that absorbs it is the restored, misaligned spread of the paragraph above in another currency. The map is fitted on the rows it is then scored on, two free parameters against a median of five solvents, so it bounds what the concession can buy rather than measuring it. Refitted out of sample it returns an interval wide enough to hold both the whole gap and none of it, and the width is fixed by the design rather than by the answer: a two-parameter map fitted on the other two rows of a three-row group is exactly determined (§S6.1).

## 3.3 What the substitution does not decide, and what would settle it

We do not resolve which of the two channels the substitution fails through. Drug-like solutes are essentially absent from the VT-2005 set, so the substitution falls almost entirely on the *solvent* channel, and the both-channel substitution the attribution would need covers ≈7 solutes. Enlarging that set means computing new reference profiles rather than looking them up.

*Why* it hurts stays open too, and for a stronger reason than a test that came back empty. The reading it invites is that handing the closure a faithful input surfaces a misspecified map instead of repairing it, so the reference profile is the better input where the map that consumes it is faithful. An account from outside chemistry would reach the same place without any of that. Wu et al.[41] substitute near-exact Reynolds stresses into the Reynolds-averaged Navier–Stokes equations, recover large velocity errors, and trace them to ill-conditioning of the fixed model. It does not carry here, and the reason is the size of the step. The reference profile stands 1.53 $[1.39, 1.80]$ times the learned profile's own norm away from it, and a random direction of that same norm amplifies *more* often than the substitution does (§S11.1). The reading is about a fixed map's own error and about what its inputs fail to carry, and on the solubility axis those two terms stay fused. The solubility axis is where §3.1 measured the substitution, so both quantities lie beyond observation on those rows. §3.4 measures a second axis and leaves the first unexplained.

Two experiments would settle it and both remain to be run. The first is the transfer back to the paradox's own regime, which is limitation (i). The second is limitation (x). The fidelity contrast of §3.4.3 compares *oracle-input* activity error across kernels, so $\mathcal{G}$ itself stands measured under no modern closure on either axis, and whether the *solubility* paradox survives a 2010/dsp closure is untested. Both need a typed $\sigma$-profile head trained end to end through the 2010 kernel, and the solubility one additionally needs a UD ∩ BigSolDB matched set. The set exists but is thin: 437 of the 5608 labelled test rows carry a UD profile on both sides, over 11 solutes and 57 pairs, so it prices a handful of solutes rather than the corpus.

## 3.4 A second axis: bounding a deployed COSMO-SAC against its inputs

Our estimator is the binning bound, read at the estimator cell §2.5 fixes in advance and reserves to this section. The cell is eight equal-count bins of the conditioning variable, the residual-only convention, and the unbiased within-bin variance, with admissibility clauses (a)–(c) deciding which strata may be read at all. We fixed all four before computing any margin, and §3.1–§3.3 are measured with other instruments. The decomposition of §2.1.1 separates a fixed closure's error into a part in the map and a part in its inputs. Its output is a map over chemical strata rather than one verdict for a set, for the reason §3.4.1 establishes. It runs on the two data sets of Table S1, which we keep apart throughout.

One reading rule governs every margin below, and it is Lemma 1's. A positive margin establishes $B_{\text{closure}} > B_{\text{insuff}}$. Its size is a *lower bound* on $B_{\text{closure}} - B_{\text{insuff}}$ and not an estimate, so an interval on a margin bounds that bound. A margin at or below zero leaves the two terms merged and licenses nothing, their equality staying open. $B_{\text{closure}}$ throughout is the error of the fixed map *read at a computed profile*, so a positive margin is a property of that composite and not of the kernel alone (§3.4.3). The one cell serves both sets, both kernels and every stratum, so every cell below was fixed before the number it returned was seen.

### 3.4.1 The broad IDAC set ($n$=477)

The closure reads a temperature as well as a profile pair, so its argument is the concatenated UD profile pair with $T$: 103 dimensions against $n$=477. The target $m$ is the experimental $\ln \gamma_2^\infty$. Two things scope the set before any number is read. Its chemistry is small — every solute has 12 heavy atoms or fewer against a median of 14 for our own drug-like solids — so *broad* names the reach of the $\gamma^\infty$ corpus the kernel is calibrated on, covering the solvent side of deployment and leaving the solute side open. And every MSE, bound and margin on this axis is in squared $\ln \gamma_2^\infty$ units, a different axis from the $\ln x_2$ one the MAEs of §3.1 are measured on; the two are not comparable.

For the set taken whole the closure scores MSE = 1.902 against Var($m$) = 4.85, the binning bound gives $B_{\text{insuff}} \lesssim 0.70$, and the margin is +0.51. That aggregate is not the output, because it is unstable under composition: dropping the two chemistries the error concentrates in takes it to −0.12. An aggregate margin is a composition-weighted average of a term free to move 25-fold against one that barely moves, so it reports a binding closure for any set whose composition includes a chemistry the closure fails on (§S3.8). The output is therefore the map, stratum by stratum, and the bound stays valid in each stratum where it is computed.

*The map, and how much of it survives the rule.* We run the instrument over 59 strata of the broad IDAC set (477 rows, 185 pairs, 78 solutes, 36 solvents, 15 source publications). Fifteen are boundable; six of those are admissible, and clause (c) makes the six three distinct row sets, of which one is both positive and left in place by its residual test. Tables S16 and S17 print all fifteen and §S3.8 the other 44, each with the test it fails. **The glycol-ether solvents are the one row set the rule leaves standing, and we report that as the instrument's output rather than as a finding of this work.** On them the closure scores MSE = 2.25 against an input-insufficiency bound of $B_{\text{insuff}} \lesssim 0.11$, hence $B_{\text{closure}} \gtrsim 2.14$: the closure's own error exceeds what input insufficiency can account for by at least +2.04, 90% interval [+1.25, +2.87]. The sign survives every single-publication deletion, every glycol deleted in turn, the loss of the alkane solutes, and every cell of the bin-count sweep, which is the analyst choice with the largest leverage here (§S3.3, §S3.9).

Three things limit what that licenses, and each is a property of the design rather than a caveat we add. *The row set is the survivor of a 59-fold search.* A chemistry-blind relabelling of the same molecules reaches its margin in 10.1% of draws, so as a chemistry-specific effect it is unestablished against the search that found it; the instrument licenses the separation on the rows it was computed on and no more (§S3.6). *Its chemistry is narrow*: 182 rows over 43 pairs, four glycols of one homologous series, 19 solutes every one a hydrocarbon or a halobenzene, and two of three publications carrying 177 of the rows. *And the separation reproduces at about half its size on rows the search never saw.* An independent ThermoML publication supplies 95 rows the broad set does not carry, where the margin is +1.06 $[+0.74, +1.29]$ against the in-sample +2.04. It carries one source publication, so the leave-one-source-out clause cannot be applied and the reading is descriptive rather than admissible. It settles the sign, and suggests the in-sample magnitude is the upper end of what this chemistry supports rather than its size.

**Deviation from the declaration.** A pre-declared cross-fitted regressor was to replace the binning bound if it passed a validity gate. On the declaration's own folds it returns no bound at all — on the glycol-ether stratum it puts $B_{\text{insuff}}$ at 4.92 against that stratum's MSE of 2.25, twice what Lemma 1(a) allows — so its outputs there are uninformative rather than reversals. **But the declared gate did not test that, and on the configuration it did test the declaration required unconditional adoption; every margin printed in this paper is nevertheless the binning bound's.** Where the cross-fit does pass it is uniformly looser and does not favour us: it puts the same excess at +1.76 $[+1.02, +2.49]$, and moves the chemistry-blind null so that a relabelling reaches the observed maximum in 54% of draws against 16% under the binning bound. Under the estimator the declaration named, this cell sits at chance. Nothing is claimed for it either way, since it is reported as a survivor and not as an established effect — but the deviation runs in this paper's favour and was chosen after the two bounds were compared, which is what a declaration exists to prevent. It is the sharpest thing a reader can hold against this work, and the poorer cross-fitted map prints cell by cell in §S3.12 so that the declared rule can be held against it.

A selection objection would be settled by the same measurement on rows the search never saw, and we ran one against another group's compilation, sixteen times the broad set's rows on a different profile database (Table S27). **The chemistry stayed untestable there**: the cell the finding is about, a glycol-ether solvent with an *organic* solute, is empty in that database. *Not testable* is the third verdict the pre-declaration lists and counts as non-confirmation, and it follows from the row inventory alone, so what the pre-registration delivers here is the absence of confirming data rather than a failed replication (§S3.7, Table S11).

#### 3.4.2 The VT-2005-matched set ($n$=60)

This set carries the paper's other reference-matched analyses and is far weaker. All 60 rows sit at 298 K, so that $\dim(z^\star)/n \approx 102/60$, and methanol occupies 25 of them. Its 17 solvents fall in six of the nine classes, each short of the $n$=40 a bound needs. The set supports no stratified bound, and we state no stratum of it as a finding. Its aggregate margin of +0.05 collapses four ways: under deletion of the two leverage pairs, under dropping methanol, under blocked out-of-fold estimators, and under 4 of the 60 single-pair deletions. The worst of those deletions leaves −0.03, where on the broad set every single-pair deletion leaves the sign intact. A margin here would in any case bound a label systematic tracking $z^\star$, which the unbiased within-bin variance leaves *open* (§S3.11). The reported cell is not a favourable one: drawn over the whole grid of bin counts and conventions the aggregate changes sign, and of the cells returning a positive margin the reported one is near the bottom (Fig. S2, §S3.3). Of the 60 pairs, 57 also appear in the broad IDAC set at 298 K under UD profiles, so the two are overlapping samples rather than independent ones. VT-2005, and therefore the paradox's own oracle, is defined on this sub-design, so the agreement between the two sets' error orderings is a consistency check rather than corroboration.

#### 3.4.3 Whether a better closure moves the ceiling, and where this one fails

If the ceiling is the closure, the direct remedy is a better closure. The standard 2002 → 2010 step delivers one on the VT-2005-matched set and withholds it on the broad IDAC set. Neither half of that pair settles anything. The improvement is on a 57-pair overlap with the larger set rather than an independent sample, and the withholding is $-0.99$ at $P \approx 0.2$, which is not a measured equality either. Three further readings are of the same kind: the 2010 kernel on its own typed latent, a fitted kernel residual asked to transfer across solvents, and the block split that places the reversal in the chemistry rather than the temperature range. Each prints with its interval in Table S9 and §S6.6. We claim only that **the fidelity lever we can demonstrate is confined to the smaller set's chemistry**, and that broad-set null is itself a small-molecule and glycol-heavy one, so on drug-like solids the lever stands untested.

On the one row set the rule leaves standing, where $B_{\text{closure}}$ dominates, the ceiling is in the map *as it is deployed*: the fixed kernel together with the one computed conformer it reads. The composite is what $B_{\text{closure}}$ is the error of, and the object the pipeline runs on. The split between the two stays open here, and so does whether the composite is *fixably* wrong. A fitted kernel residual on the 60 matched pairs transfers within a solvent and fails to transfer across solvents, under the solvent-grouped folds that are the one fold design free of leakage. The result is a null. So $B_{\text{closure}}$ is an error a handful of kernel parameters can absorb *within* a solvent, and its correctability across solvents we leave unshown.

The failure is chemically placed. On the VT-2005-matched set 90% of the squared error falls on strong hydrogen-bond-donor solvents, and the closure is near-exact on acceptor-only ones. Those are the chemistry the documented single-parameter H-bond term governs, and the term the donor/acceptor-typed 2010 closure was built to replace.[32,37] That failure is established already and the localisation restates it. The new part is that on one such chemistry the failure is bounded against what the closure's inputs leave unresolved. The bound also survives the obvious alternative: replacing the four glycol-ether references with Boltzmann ensembles moves the stratum's margin from $+2.04$ to $+2.27$, so the error sits in a *chemistry* rather than in the conformer half of the composite (§S3.10).

COSMO-SAC-2002 being a low-fidelity closure is the contribution and not a qualification of the result. It is what a practitioner adopts by default, and a closure's fidelity is *unknowable in advance* on a new system. The field routinely composes an expressive encoder with a fixed mechanistic map and assumes the map is good enough. Once that assumption fails, the learned latent silently departs from the quantity it is named after (§S6.10), and an external oracle is what makes the departure visible. Outside that one row set the map reports cells it can only leave unresolved, which is a weaker statement than cells in which the closure is sound. The exposure the choice of profile database creates, and why that arithmetic sizes it while leaving its share unbounded, is in §S6.6.

## 3.5 The same split on a second closure: pKa through a fixed Hammett relation

The instrument generalises beyond COSMO-SAC, and a second domain runs it where the sign of grounding can be predicted before the measurement. The closure is a Hammett linear free-energy relationship $g(z^\star) = \mathrm{p}K_{a,0} - \rho\,\sigma^\star_{\mathrm{H}}$ in tabulated Hansch–Leo–Taft constants,[42] unfitted on these data, so $g$ is fixed in the sense Lemma 1 requires. The learned quantity is the substituent constant predicted from structure, and the data are the OPERA compilation.[43] A rule reading structure alone splits the records into two poles. The *clean* pole is the 158 whose every substituted position is meta or para with a tabulated substituent. The *degraded* pole is the 846 where $\sigma^\star_{\mathrm{H}}$ is a partial sum over the positions the table can see.

The sign reverses between them, in the direction that construction fixes in advance. Across $K{=}20$ within-scaffold splits the reference *improves* accuracy on the clean pole (MAE 0.48 learned against 0.315, a gap of $-0.165$ at a 90% confidence interval of $[-0.28, -0.04]$) and *degrades* it on the degraded pole (0.83 against 1.62, $+0.79$ $[+0.57, +0.95]$). The freely learned constant absorbs what the zeroed positions carry and the tabulated one lacks. **The flip is an in-distribution result and the scaffold protocol does not certify it.** Re-run on a scaffold split the reference helps on both poles, so the degraded pole's "grounding hurts" half fails to certify and so does the reversal. The scaffold run confirms the mechanism, not the flip. We measure every solubility result in this paper on a scaffold split, and this is the one place the protocol changes. §S6.12 carries the assignment rule, the bounds on both poles, what the quality filter decides, the estimator's two conditions of use, the competence sweep and the $G$=4 sign test.

### 3.6 Scope, accuracy, and limitations

Two measurements, in different places. On solubility, supervising a learned physical latent on reference values during training moves the error without an established sign, while *supplying* those values at inference hurts at every seed. On infinite-dilution activity data, and there on one class of solvent, the fixed map's own error exceeds the bound on what its inputs leave undetermined. The two sit in different regimes on largely disjoint chemistry, so "the substitution hurts *because* the map is misspecified" remains untested (limitation (i)). The finding is one-sided: one row set of the map on which grounding effort is *not* the remedy.

**The unconstrained control's representation.** If the physics arm's intermediate is a surrogate, the thermodynamics might still be present implicitly in the control, which is free to represent it. It is absent: a ridge probe recovers the control's own prediction at $R^2 = 0.98$ and leaves the measured ideal-solubility term unrecovered. The negative falls short of an exclusion, on 13 held-out solute clusters with no power calculation behind them; it is limitation (ix), and §S9 carries the three studies and their controls.

**Where the accuracy stands, and the confound in it.** None of this work's arms is competitive with current solubility models, and none is offered as one; the comparison in Table S30 fixes the scale this work operates on and does not rank anything. The substitution result is a contrast within one checkpoint and does not depend on that scale. The physics arms also trail their own control in the seed mean (Table S4), but the two sides are separately tuned with no hyperparameter-matched control, so no difference between them, and none of the chemical strata that resolve the same difference (§S6.4, §S6.5), is attributable to the thermodynamic bottleneck. We claim no accuracy result in either direction, and the matched control that would repair this is unrun. Untouched by that confound is every *within-checkpoint* contrast, where the two sides share one trained model and one tuning (§3.1, §3.2, §S6.10, §3.4). Two fall outside it, comparing across trainings or across the separately tuned control: the supervision gain and §3.2's shrinkage slopes. Grounding the crystal factor on external labels is underpowered rather than null: 97.76% of that pool was melting points the model had already been trained on, and the downstream solubility gap crosses zero once clustered by solute (§S6.3). Temperature extrapolation is the one place the physical *structure* helps, and it helps as a property of the hypothesis class rather than of any arm trained here (§S6.1).

The external block is on an easier by-solute split, and the harness for a like-for-like comparison is deposited and unrun. The difference that remains is a scale rather than a placement. This work's control reaches $\log_{10} S$ RMSE 1.00 on held-out scaffolds and its physics arms 1.44 to 1.64, against 0.83 to 0.99 for the two leading direct models and 1.43 to 2.16 for the physics-informed SolProp cycle.[2] All six are as published by Attia et al.,[1] in $\log_{10} S$ RMSE in $\mathrm{mol\,L^{-1}}$, on two other corpora: the SolProp test set, whose solutes are disjoint from its training corpus, and the near-ambient Leeds set of Boobier et al.[44] Sharing no row with these, they set a scale and withhold an ordering; each range is two point values on two corpora rather than a spread, and none of the six carries a published uncertainty. The clip sweep, the random-pair rows and the conversion behind the $\log_{10} S$ column are in §S4.

These negatives fall where Remark 1 predicts a prior will hurt (§S6.9). The synthetic sweep of Fig. S12 is true by construction there, so it verifies the instrument and not the hypothesis; the pKa closure exercises the sufficiency channel on real data.

**Limitations.** All sixteen, keeping the numerals used throughout; §S6.8 carries the qualifying detail for the nine it holds.

*What the fixed model leaves out.* Properties of the closure and of what it is asked to do, not of the measurement.

(xvi) The fixed model is COSMO-SAC at its 2002 parameterisation, whose kernel cannot move, so a sign that turned on kernel mobility lies outside what these measurements reach (§S6.8).

(xv) Two terms the thermodynamics asks for are set to zero in the deployed model, $\Delta C_p$ in the ideal term and the Staverman–Guggenheim combinatorial term, and both are absorbed at training time by the same learned $\sigma$ head (§S6.8).

(xiv) One trained COSMO-SAC head survives in the deposit and it is a poor model, so the four quantities read off it are properties of a parameterisation rather than of a trained network, each stated with that scope where it prints (§S6.8).

(iv) $B_{\mathrm{insuff}}$ is upper-bounded rather than point-identified, the design holding no replicates at fixed $z^{\star}$. $B_{\mathrm{closure}}$ is the error of a composite, the fixed map together with the calculation that produced $z^{\star}$, and the measurements here leave it undivided between the two and its share unbounded (§3.4.3).

*What the instrument can and cannot resolve.* The estimator, its analyst choices and the multiplicity the search carries.

(ii) What the convention rule costs the aggregate (§2.5).

(iii) That other cells of the map keep their sign as well (§3.4.1).

(v) The constant-offset bound's convention-specificity (§2.5).

(vi) The stratification reads structure alone, so every cell was fixed before its outcome was seen; cells from different axes share rows and are counted once, and the per-cell bootstrap frequencies carry no multiplicity control (§S6.8).

(x) What the closure-fidelity contrast leaves unmeasured, and what would settle it (§3.3).

*What the data and the runs bound.* Provenance, scale and the ceiling the labels themselves impose.

(xii) The scaffold-leak guard is a property of the released stream builder rather than a certified property of the runs §3.1 reports; the certified leak-free re-run returns the supervision gain without a sign, so no reported number rests on the uncertified build (§S6.8).

(vii) The accuracy ceiling standing a factor of five above the label-noise floor (§2.2).

(xiii) What the deposit leaves uncheckable (the data-availability statement).

(viii) What the three-seed latent-departure result establishes and where it stops (§S6.10).

(ix) What bounds the black-box probe (the probe paragraph above).

*How far any of it generalises.* The two limits on carrying these results past the setting they were measured in.

(i) The phenomenon and its causal attribution lie in *different regimes*: the paradox is where the model operates, the attribution rests on infinite-dilution activity data alone, and transfer between them is a conjecture and the principal open question here (§S6.8).

(xi) No accuracy difference between the arms is attributable to the bottleneck, the two sides being separately tuned with no hyperparameter-matched control (§S6.8).

# 4 Conclusions

In this work we tested whether supplying a physics-informed solubility model with tabulated values for its own physical intermediate improves it, and found that grounding is not one operation but two, with divergent outcomes. Substituting reference $\sigma$-profiles into a fixed COSMO-SAC closure at prediction time *degraded* accuracy: over five seeds the mean absolute error rose from 1.93 to 2.34 in $\ln x_2$, about 21%, with no per-seed overlap between the two arms, and the mean per-solute Spearman correlation over solvents fell by 0.185 — a loss in the ranking the model is used to produce. Supervising the same intermediate against the same reference during training gave no consistent improvement, its gains straddling zero. The operation that failed is the one available in practice, because it is the one that needs no retraining.

For a fixed-closure architecture, this says the binding constraint is often the misspecification of the physical map rather than the quality of the input. Where the closure's own error dominates, a more accurate input exposes that error instead of correcting it. The effect is chemistry-specific: we resolve it on glycol-ether solvents and not on the other classes the same instrument bounds, so whether grounding helps depends on the chemical regime and has to be asked one regime at a time.

Our result differs from work reporting gains from a frozen, externally supervised intermediate,[15] and the two are not in contradiction: that construction leaves the closure's kernel free to move and ours does not, and the chemical scopes differ. What follows for practice is that external reference data are best treated as a diagnostic for locating a model's limits rather than as a remedy that can be assumed to work. The measurement that would settle the mechanism is a laboratory one — the fixed model's own target and the downstream property on the same pairs, which no public database currently supplies. Until then, freezing a learned intermediate against its reference and fine-tuning only downstream is the safer construction, revalidated for each chemical domain before it is relied on.

# Author contributions

N. L. Polomoshnov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. A. V. Rudik: Writing – review & editing.


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# Notes

The authors declare no competing financial interest.

# Data and Software Availability

The analysis code, and those intermediate artifacts small enough to version, are available at `https://github.com/doctawho42/tgnn-solv` (commit `fb568bca64`, tag `v1.0.0`) under the MIT licence, that commit being the one the reported numbers were produced from. The model checkpoints, the processed split, the full per-arm prediction files and the larger analysis artifacts will be deposited in a Zenodo archive, **[whose DOI is registered at submission]**. The data sources are public. Solubility is BigSolDB 2.0;[34] the reference $\sigma$-profiles are the VT-2005 database, and the UD profiles Ref.[32] as redistributed with Ref.;[33] the infinite-dilution activity coefficients are ThermoML, with the PGL 6th-edition IDAC database behind Table S27; the pKa data are the OPERA compilation.[43] Four disclosed defects sit on the deposited record, in §S8 and §S3. The three-seed surrogate run's per-run split provenance was not logged; one $\sigma$-stream build on disk was written against the wrong split files; the two VT-2005-matched audit artifacts disagree on one out-of-fold estimate. The fourth is that the $\sigma$-stream those runs consumed is recorded by digest in every checkpoint manifest and was not itself retained. Rebuilding it from the recorded parameters reproduces the pool exactly, the same 1319 molecules with bit-equal profiles, and the train/validation split sizes exactly, but no retained file settles the assignment between them. That comparison is itself deposited, digest by digest, and regenerates from the released code. Every number reported here was measured on arms whose weights, predictions and manifests are deposited; what the loss bounds is what can be added to them, since the validation half early-stops the $\sigma$ warm-up. Two further defects are repaired rather than disclosed, each described where its numbers are printed. We re-scored the two FastSolv retraining bundles (Tables S30 and S20, with the diagnosis, the withdrawn values and the control that reproduces them in §S4), and the withdrawn bundle is kept beside the re-scored one. And the truncated seed-44 $\sigma$-oracle per-row file has been recovered complete and now stands in the deposit (Table S29, §S8). *The audit in one place.* Twelve defects and limits are disclosed in this submission, each described where its numbers print. §S8 lists all twelve in one place, of five kinds: four properties of the deposit, disclosed rather than repaired; two errors in this work's own numbers, repaired with the withdrawn values kept beside the corrected ones; three cases where a check this work built refuted or weakened something this work had claimed; two properties of the model, disclosed and then bounded; and one limit of the checking apparatus itself. None of the twelve is reported by anyone but us.

## Supporting Information Available

Supplementary methods; proofs of Lemma 1 and Remark 1 and three further results; per-arm tables and decomposition robustness; the map's 59 strata in full, including the 44 the estimator cannot bound; further external ablations; supplementary mechanism results; supplementary diagnostics and broad negatives; the second-domain pKa construction; data provenance and reproducibility; the black-box probe tables; the synthetic closure-fidelity family; the positioning against adjacent literatures (PDF). Broad IDAC set, 477 rows, Data Set S1 (CSV). VT-2005-matched set, 60 rows, Data Set S2 (CSV).

The Supporting Information is a separate document. Its sections, tables, figures and equations are numbered S1, S2, ..., and every reference above that carries an S is to it; the columns of both data sets are listed in §S8.

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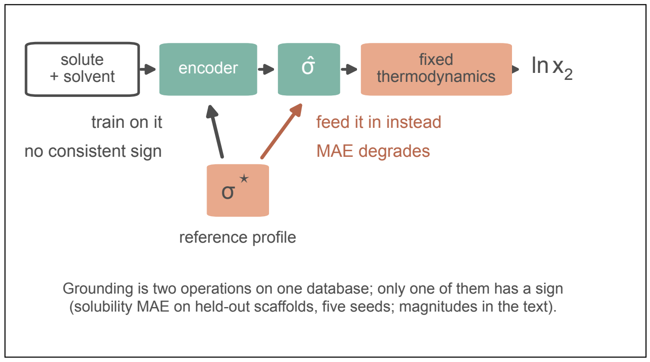
# TOC Graphic


solute
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train on it
no consistent sign
feed it in instead
MAE degrades
$\sigma^{\star}$
reference profile
Grounding is two operations on one database; only one of them has a sign
(solubility MAE on held-out scaffolds, five seeds; magnitudes in the text).